

Part A: Theoretical Foundation

1. What is Inferential Statistics?

Inferential statistics involves using data from a sample to make inferences, predictions, or generalizations about a larger population. It allows us to draw conclusions beyond the data actually collected by calculating probabilities and estimating population parameters (like the mean) based on sample statistics.

2. What is Hypothesis Testing and its components?

It is a formal statistical method used to evaluate a claim about a population parameter.

- **Null Hypothesis (H_0):** The default assumption that there is no effect or no difference.
- **Alternative Hypothesis (H_1):** The claim we are testing for, suggesting there *is* an effect or difference.

3. Confidence Interval and Critical Value

- **Confidence Interval (CI):** A range of values, derived from sample data, that is likely to contain the true population parameter with a certain level of confidence (e.g., 95%).
- **Critical Value:** The threshold value on the distribution curve that separates the "rejection region" from the "non-rejection region." If the test statistic exceeds this, we reject H_0 .

4. Define p-value

The p-value is the probability of obtaining test results at least as extreme as the observed results, assuming the null hypothesis is true. A small p-value (typically $\leq \alpha$) indicates strong evidence against the null hypothesis, leading us to reject it.

5. Differentiate Type I and Type II errors

- **Type I Error (False Positive):** Rejecting the null hypothesis when it is actually true.
- **Type II Error (False Negative):** Failing to reject the null hypothesis when it is actually false.

6. Statistical Tests Summary

- **z-test:** Used for comparing means when the population standard deviation is known and the sample size is large ($n \geq 30$).
- **t-test:** Used for comparing means when the population standard deviation is unknown or the sample size is small.
- **Chi-Square Test (χ^2):** Used to determine if there is a significant association between two categorical variables.
- **ANOVA (Analysis of Variance):** Used to compare the means of **three or more** independent groups to see if at least one is significantly different.

7. What is Covariance?

Covariance measures the direction of the relationship between two variables. A positive covariance indicates they tend to move in the same direction, while negative covariance indicates they move in opposite directions. However, it does not indicate the *strength* of the relationship because it is not scaled.

8. What is Correlation?

Correlation (specifically Pearson's r) is a scaled measure of the linear relationship between two variables, ranging from -1 to +1. Unlike covariance, it tells us both the direction and the strength of the relationship, regardless of the units of measurement.